

practical-5

Aim: Implementation of a knapsack problem using dynamic programming.

Knapsack:

Algorithm:

Algorithm:-
Input:- Number of items n , knapsack capacity w , weights $w[i]$, and values $v[i]$
Output:- Maximum value and selected items.
Step1:- Start
Step2:- Read $n, w, w[i]$, and $v[i]$
Step3:- Initialize $dp[i][0] = 0$ and $dp[0][j] = 0$
Step4:- for $i = 1$ to n and $j = 1$ to w , repeat
Step5:- if $w[i] \leq j$,
 $dp[i][j] = \max(v[i] + dp[i-1][j-w[i]], dp[i-1][j])$
Step6:- otherwise, $dp[i][j] = dp[i-1][j]$
Step7:- Display $dp[n][w]$ as maximum value.
Step8:- Trace the table to find selected items.
Step9:- Stop

Program:

```
#include<iostream>
using namespace std;

class Knapsack {

private:
```

```
int w[10];    // weights
int v[10];    // values
int n;        // number of items
int W;        // capacity
int dp[10][100]; // dp table
```

public:

```
// — Get Input —
void getInput() {
    cout << "Enter number of items : ";
    cin >> n;
    cout << "Enter knapsack capacity : ";
    cin >> W;
    for(int i = 1; i <= n; i++) {
        cout << "Item " << i << " weight and value : ";
        cin >> w[i] >> v[i];
    }
}
```

```
// — Build dp table —
void solve() {

    // base case — row 0 and col 0
    for(int i = 0; i <= n; i++) dp[i][0] = 0;
    for(int j = 0; j <= W; j++) dp[0][j] = 0;
```

```
    // fill table
    for(int i = 1; i <= n; i++) {
        for(int j = 1; j <= W; j++) {
```

```
            if(w[i] <= j) {
                // take or not take
                dp[i][j] = max(
                    v[i] + dp[i-1][j-w[i]],
                    dp[i-1][j]
                );
            } else {
                // cannot take item i
                dp[i][j] = dp[i-1][j];
            }
        }
    }
```

```
    }  
    }  
}  
  
// — Print dp table —  
void printTable() {  
    cout << "\nDP Table:" << endl;  
    cout << "    ";  
    for(int j = 0; j <= W; j++)  
        cout << j << " ";  
    cout << endl;  
  
    for(int i = 0; i <= n; i++) {  
        cout << "i=" << i << " ";  
        for(int j = 0; j <= W; j++)  
            cout << dp[i][j] << " ";  
        cout << endl;  
    }  
}  
  
// — Find selected items —  
void findItems() {  
    cout << "\nSelected Items:" << endl;  
    int j = W;  
    for(int i = n; i >= 1; i--) {  
        if(dp[i][j] != dp[i-1][j]) {  
            cout << "Item " << i  
                << " (w=" << w[i]  
                << ", v=" << v[i]  
                << ") TAKEN " << endl;  
            j = j - w[i];  
        } else {  
            cout << "Item " << i  
                << " (w=" << w[i]  
                << ", v=" << v[i]  
                << ") NOT taken " << endl;  
        }  
    }  
}
```



```
// — Show result —  
void showResult() {  
    cout << "\nMaximum Value = "  
        << dp[n][W] << endl;  
}  
};  
  
// — MAIN —  
int main() {  
  
    Knapsack k;  
  
    k.getInput();  
    k.solve();  
    k.printTable();  
    k.showResult();  
    k.findItems();  
  
    return 0;  
}
```

Output:



Output

```
Enter number of items : 4
Enter knapsack capacity : 5
Item 1 weight and value : 6 7
Item 2 weight and value : 8 3
Item 3 weight and value : 9 2
Item 4 weight and value : 7 9
```

DP Table:

	0	1	2	3	4	5
i=0	0	0	0	0	0	0
i=1	0	0	0	0	0	0
i=2	0	0	0	0	0	0
i=3	0	0	0	0	0	0
i=4	0	0	0	0	0	0

Maximum Value = 0

Selected Items:

```
Item 4 (w=7, v=9) NOT taken
Item 3 (w=9, v=2) NOT taken
Item 2 (w=8, v=3) NOT taken
Item 1 (w=6, v=7) NOT taken
```

=== Code Execution Successful ===

Time Complexity:

Best Case:



Time Complexity:-

Best Case:-

```
for i = 1 to n  
  for j = 1 to w  
    dp[i][j]..
```

$$\begin{aligned}T(n) &= n \times w \\ &= nw \\ T(n) &= O(nw)\end{aligned}$$

Average Case:

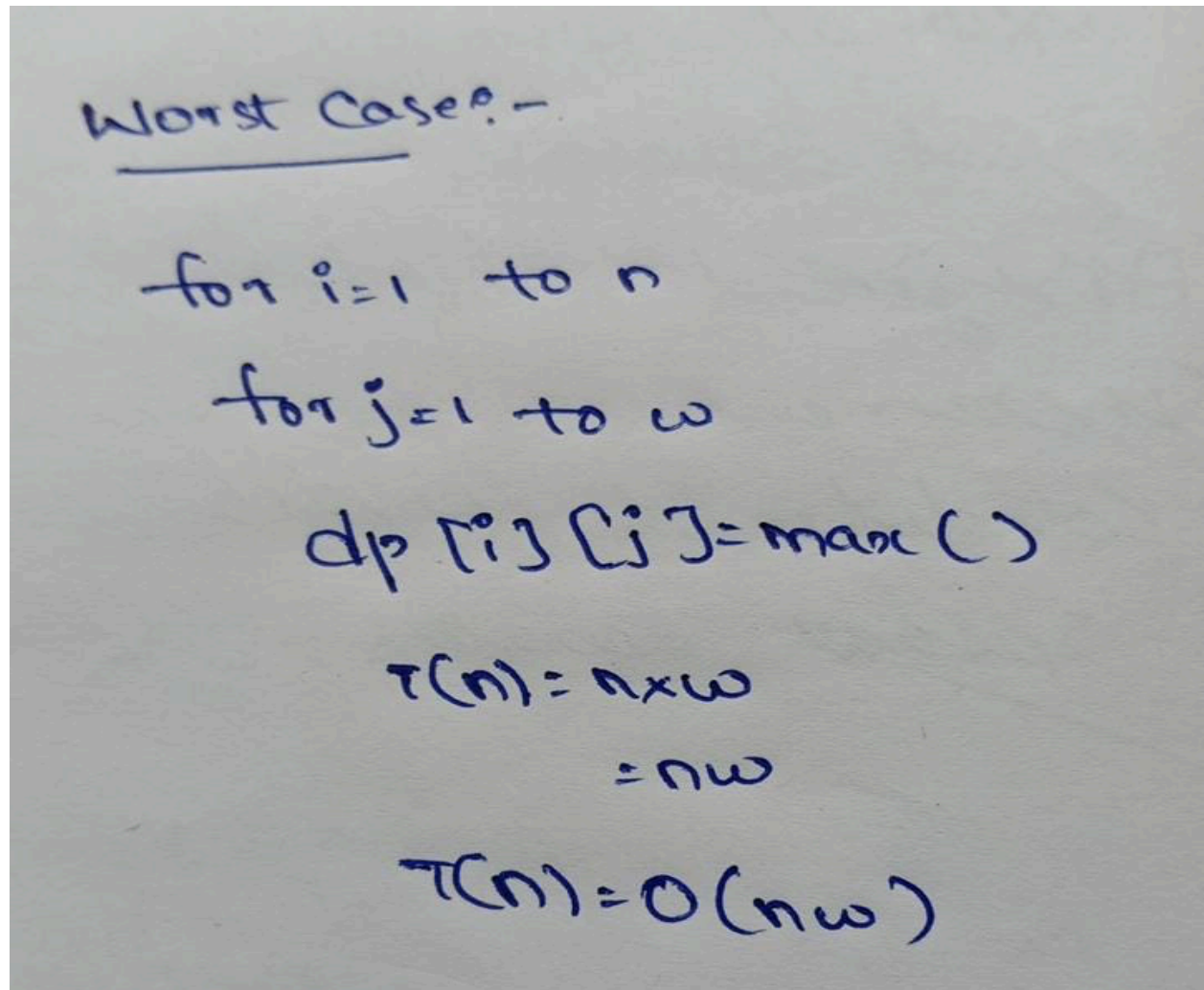
Average case:-

```
for i = 1 to n  
  for j = 1 to w  
    if (w[i] <= j)  
      dp[i][j] = max()  
    else  
      dp[i][j] = dp[i-1][j]
```

$$\begin{aligned}T(n) &= n \times w \\ &= nw \\ T(n) &= O(nw)\end{aligned}$$



Worst Case:



Conclusion:

The implementation and time analysis of 0/1 Knapsack using Dynamic Programming helps us understand its efficiency. The algorithm builds a DP table and considers whether to include or exclude each item to find the maximum value within the given capacity. It has a time complexity of $O(nW)$ in the best, average, and worst cases. Therefore, Dynamic Programming provides an efficient and consistent solution for solving the 0/1 Knapsack problem.



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Design and analysis of algorithm – 01AI0506